

ALFAAB24648

## Diallyl phthalate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>邻苯二甲酸二烯丙酯</b><br><b>Diallyl phthalate</b>                                                                                                                                                                                                                       |
| <b>Cat No. :</b>                                      | <b>B24648</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                                       | 1,2-Benzenedicarboxylic acid di-2-propenylester                                                                                                                                                                                                                    |
| <b>CAS No</b>                                         | 131-17-9                                                                                                                                                                                                                                                           |
| <b>Molecular Formula</b>                              | C14 H14 O4                                                                                                                                                                                                                                                         |
| <b>Supplier</b>                                       | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                                            |                                   |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                                                                                                                                                            | <b>Appearance</b><br>Light yellow | <b>Odor</b><br>Slight |
| <b>Emergency Overview</b><br>Harmful if swallowed. May be harmful in contact with skin. Very toxic to aquatic life with long lasting effects. Lachrymator (substance which increases the flow of tears). May cause an allergic skin reaction. Harmful if inhaled. Suspected of causing genetic defects. May cause damage to organs through prolonged or repeated exposure. |                                   |                       |

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 4 |
| Acute Dermal Toxicity                                | Category 5 |
| Acute Inhalation Toxicity - Vapors                   | Category 4 |
| Skin Sensitization                                   | Category 1 |
| Germ Cell Mutagenicity                               | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Acute aquatic toxicity                               | Category 1 |
| Chronic aquatic toxicity                             | Category 1 |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H313 - May be harmful in contact with skin  
H410 - Very toxic to aquatic life with long lasting effects  
H317 - May cause an allergic skin reaction  
H341 - Suspected of causing genetic defects  
H373 - May cause damage to organs through prolonged or repeated exposure  
H302 + H332 - Harmful if swallowed or if inhaled

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P202 - Do not handle until all safety precautions have been read and understood  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Harmful if swallowed. May be harmful in contact with skin. May cause an allergic skin reaction. Harmful if inhaled. Suspected of causing genetic defects. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Very toxic to aquatic life with long lasting effects. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No   | Weight % |
|-------------------|----------|----------|
| Diallyl phthalate | 131-17-9 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration.

**Ingestion**

Clean mouth with water. Get medical attention.

**Most important symptoms and effects**

May cause allergic skin reaction. . Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Prevent product from entering drains. Keep in suitable, closed containers for disposal. Do not flush into surface water or sanitary sewer system. Do not let this chemical enter the environment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE**

**Handling**

Avoid contact with skin and eyes. Do not breathe mist/vapors/spray.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component         | ACGIH TLV | OSHA PEL | NIOSH | The United Kingdom                                                 | European Union |
|-------------------|-----------|----------|-------|--------------------------------------------------------------------|----------------|
| Diallyl phthalate |           |          |       | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr |                |

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

## Diallyl phthalate

|                                 |                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                            |
| Environmental exposure controls | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained. |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                               |                                   |
|-----------------------------------------|-------------------------------|-----------------------------------|
| Appearance                              | Light yellow                  |                                   |
| Physical State                          | Liquid                        |                                   |
| Odor                                    | Slight                        |                                   |
| Odor Threshold                          | No data available             |                                   |
| pH                                      | No information available      |                                   |
| Melting Point/Range                     | -70 °C / -94 °F               |                                   |
| Softening Point                         | No data available             |                                   |
| Boiling Point/Range                     | 300 - 167 °C / 572 - 332.6 °F | @ 5 mmHg                          |
| Flash Point                             | 166 °C / 330.8 °F             | Method - No information available |
| Evaporation Rate                        | No data available             |                                   |
| Flammability (solid,gas)                | Not applicable                | Liquid                            |
| Explosion Limits                        | No data available             |                                   |
| Vapor Pressure                          | 4 mmHg @ 160 °C               |                                   |
| Vapor Density                           | 8.3                           | (Air = 1.0)                       |
| Specific Gravity / Density              | 1.120                         |                                   |
| Bulk Density                            | Not applicable                | Liquid                            |
| Water Solubility                        | 148 mg/L (20°C)               |                                   |
| Solubility in other solvents            | No information available      |                                   |
| Partition Coefficient (n-octanol/water) |                               |                                   |
| Component                               | log Pow                       |                                   |
| Diallyl phthalate                       | 3.23                          |                                   |
| Autoignition Temperature                | 435 °C                        |                                   |
| Decomposition Temperature               | No data available             |                                   |
| Viscosity                               | 13 mPa.s at 20 °C             |                                   |
| Explosive Properties                    | No information available      |                                   |
| Oxidizing Properties                    | No information available      |                                   |
| Molecular Formula                       | C14 H14 O4                    |                                   |
| Molecular Weight                        | 246.26                        |                                   |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                      |
|--------------------------|------------------------------------------------------|
| Stability                | Stable under normal conditions.                      |
| Hazardous Reactions      | No information available.                            |
| Hazardous Polymerization | No information available.                            |
| Conditions to Avoid      | Incompatible products.                               |
| Materials to avoid       | Strong bases. Strong acids. Strong oxidizing agents. |

Hazardous Decomposition Products Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|-----------|-------------|-----------------|
|-----------|-----------|-------------|-----------------|

**SAFETY DATA SHEET****Diallyl phthalate**

|                   |                          |                              |                                           |
|-------------------|--------------------------|------------------------------|-------------------------------------------|
| Diallyl phthalate | LD50 = 656 mg/kg ( Rat ) | LD50 = 3400 mg/kg ( Rabbit ) | LC50 = 5200 mg/m <sup>3</sup> ( Rat ) 1 h |
|-------------------|--------------------------|------------------------------|-------------------------------------------|

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;

Respiratory

Skin

Based on available data, the classification criteria are not met  
Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity;

Category 2

(f) carcinogenicity;

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

(i) STOT-repeated exposure;

Category 2

Target Organs

No information available.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Other Adverse Effects

See actual entry in RTECS for complete information

Symptoms / effects, both acute and delayed

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component         | Freshwater Fish                              | Water Flea         | Freshwater Algae | Microtox |
|-------------------|----------------------------------------------|--------------------|------------------|----------|
| Diallyl phthalate | LC50: 0.23 mg/L/96h<br>(Oncorhynchus mykiss) | EC50: 5.5 mg/L/48h |                  |          |

Persistence and Degradability

Persistence

Degradation in sewage treatment plant

Readily biodegradable

Soluble in water, Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative Potential

Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Diallyl phthalate | 3.23    | No data available             |

## Diallyl phthalate

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Diallyl phthalate  
**Hazard Class** 9  
**Packing Group** III

IMDG/IMO

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Diallyl phthalate  
**Hazard Class** 9  
**Packing Group** III

IATA

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Diallyl phthalate  
**Hazard Class** 9  
**Packing Group** III

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|--------|------|-----|-------|------|------|------|------|
|           |                                                     |                                         |      |       |        |      |     |       |      |      |      |      |

## Diallyl phthalate

|                   |   |   |   |   |           |   |   |   |   |   |   |          |
|-------------------|---|---|---|---|-----------|---|---|---|---|---|---|----------|
| Diallyl phthalate | - | - | X | X | 205-016-3 | X | X | X | X | X | X | KE-02288 |
|-------------------|---|---|---|---|-----------|---|---|---|---|---|---|----------|

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 26-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**  
 Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**